



Teheran University of Medical Sciences

International campus Vice-Dean for Educational Affairs

No. 124, Baradaran-e-Mozafar St.
Dameshgh St., Vali-e-Asr Ave.
Tehran 3439123900 Iran.
Tell: (+98) 21-88 89 36 78
Web: gsia.tums.ac.ir/en/icedu
Email: icedu@tums.ac.ir

Histology: Dr. Rastegar

Histology		Question time (seconds): →
1	In which type of epithelial tissue's junctions are found cadherin or claudin respectively?	
1	Desmosomes- Tight	
2	Tight- hemidesmosomes	
3	Gap- desmosomes	
4	Zonula- Gap	

Histology		Cell histology
2	Regarding the structures of the apical surface of the epithelial cells, which item is not relevant?	
1	microvilli - actin	
2	cilia - dynein	
3	microvilli - microtubule	
4	stereocilia - actin	

Histology		Question time (seconds): →
3	Regarding the basement membrane of the epithelial cell, all the items are correct EXCEPT:	
1	It has a combination of macromolecules ✓	
2	It consists of two dense and light layers	
3	Type IV collagen exists in a network.	
4	It plays a role in connecting the cell to the underlying tissue	

Histology		
4	What is the Metaplasia in epithelial tissue?	
1	the neoplastic changes	
2	the irreversible change of one type to another type	
3	the reversible changes of some cells	
4	the reversible change of one type to another type	

		Histology	Question time (seconds): →	
5		What is the following is INCORRECT?		
	1	Striation was seen in longitudinal section of cardiac muscle and skeletal muscle ✓		
	2	cardiac muscle and smooth muscle nucleus located in center part		
	3	cardiac muscle and skeletal muscle have sarcomere		
	4	Striation was seen in transverse section of cardiac muscle and skeletal muscle		

		Histology	Question time (seconds): →	
6		In the structure of the sarcomere.....		
	1	Myosin is a contractile protein in the M band		
	2	Troponin is located in A band		
	3	Transverse Tubules are located between A and I band		
	4	Titin is one of the thin filaments.		

		Histology	Question time (seconds): →	
7		In muscle tissue.....		
	1	In smooth muscle, troponin has three subunits.		
	2	Satellite cell is seen in endomysium		
	3	Only myosin can be seen in A band of striated muscle.		
	4	The intercalated disc is seen in the transverse section of the cardiac muscle ✓		

		Histology	Question time (seconds): →	
8		Which type of regulatory protein located in M or Z line of the skeletal muscle?		
	1	Myomesin- α actinin ✓		
	2	Actinin - Myosin		
	3	Titin- Desmin		
	4	Neboulin- Actin		

		Histology	Question time (seconds): →	
9		Which of the following cells is involved in Blood- Brain-Barrier?		
	1	Astrocytes- endothelial cells ✓		
	2	Oligodendrocytes-Satellite		
	3	Ependymal cells-Microglia		
	4	Schwann cells- Perimysium		

	Histology	Question time (seconds): →
10	Nissl bodies in the perikaryon of neurons are composed of:	
	1 Ribosomes and Lysosomes	
	2 Ribosomes and RER	✓
	3 Golgi and RER	
	4 SER and Lysosomes	

	Histology	Question time (seconds): →
11	Which organelle is abundant in axon terminals?	
	1 Ribosome	
	2 Lysosome	
	3 Golgi complex	
	4 Mitochondria	

	Histology	Question time (seconds): →
12	All of these are located in the chemical synapses, except:	
	1 Synaptic vesicles	
	2 Mitochondria	
	3 Gap junctions	✓
	4 Postsynaptic receptors	

	Histology	Question time (seconds): →
13	Spinal central canal lined by:	
	1 Ependymal cell	
	2 Oligodendrocyte	✓
	3 Schwann cell	
	4 Microglia	

	Histology	Question time (seconds): →
14	This type of neuron consist of many dendrites and one true axon, and are more:	
	1 Bipolar	
	2 Unipolar or pseudo unipolar	
	3 Multipolar	✓
	4 Apolar	

	Histology	Question time (seconds): →
15	Which of the following terms is applied to collection of neuron cell bodies (Soma) in the PNS and CNS respectively?	
1	Nucleus- Gray matter	
2	White matter- Ganglion	
3	Gray matter- White matter	X
4	Ganglion- Nucleus	

	Histology	Question time (seconds): →
16	About Unmyelinated PNS axons, which choice is CORRECT?	
1	Schwann cell starts to envelop one piece of axon	
2	Schwann cell starts to envelop multiple axons	✓
3	Oligodendrocyte cell starts to envelop multiple axons	
4	Oligodendrocyte cell starts to envelop one axon	

	Histology	Question time (seconds): →
17	Which layer of the epidermis contains cells with no nuclei or organelles?	
1	granulosum	
2	spinosum	
3	basale	
4	corneum	

	Histology	Question time (seconds): →
18	What type of glands are the ceruminous glands in ear?	
1	Eccrine gland	
2	Apocrine gland	
3	Endocrine gland	
4	Sebaceous glands	

	Histology	Question time (seconds): →
19	Which layer of the epidermis has cells which have keratohyaline granules?	
1	Stratum basale	
2	Stratum spinosum	
3	Stratum granulosum	✓
4	Stratum corneum	

	Histology	Cell histology
20	Langerhans and Merkel cells are seen respectively in which of the layers of the epidermis?	
	1	Stratum spinosum- Stratum basale
	2	Stratum basale- Stratum granulosum
	3	Stratum spinosum -Stratum granulosum
	✗	Stratum corneum- Stratum spinosum

	Histology	Question time (seconds): →
21	In which part of the Matrix Cartilage tissue is observed Chondroblast?	
	1	Surface Of Perichondrium
	2	Middle Of Perichondrium
	3	Lower Of Perichondrium
	4	Below of Perichondrium

	Histology	Cell histology
22	In which Type of the Cartilage Tissue is observed Collagen Fiber II?	
	1	Hyaline ✓
	2	Fibros
	3	Elastic
	4	Nothing

	Histology	Question time (seconds): →
23	Which of the following Histological Structure Is Observed Regular Collagen Fibers?	
	1	Perichondrium Tissue
	2	Cartilage Tissue Matrix
	3	Matrix Tissue
	4	Articular Cartilage

	Histology	Question time (seconds): →
24	Hydroxyapatite crystals near which cell has a natural deposition and by which ion is involved in cellular responses?	
	1	Osteoblast - H^+
	2	Osteocyte - Ca^{2+}
	3	Mesenchymal - H^+
	4	Fibroblast - Ca^{2+}

Histology

Question time (seconds): →

25

In which part of the Unmatur bone is observed Osteoclast?

- 1 Epiphysis Plate
- 2 Periosteum
- 3 Interstitial Tissue
- 4 End Plate Bone

Histology

Cell histology

26

In which Type of the Bone Tissue Isnot observed Haversian System?

- 1 Compact Bone
- 2 Alveolar Bone
- 3 Spongy Bone
- 4 Mature Bone

Histology

27

In which Type of the Joint Tissue is observed Synovial Cells?

- 1 Syndesmoses Joint
- 2 Diarthrosis Joint
- 3 Synostosis Joint
- 4 Sychondrosis Joint

Histology

Cell histology

28

Which Part Of Histological Structure Of Joint Tissue Is Formed F And M Cells?

- 1 Synovial Cavity
- 2 Synovial Fu lied
- 3 Nothing
- 4 Synovial Membrane

Histology

29

Russel bodies are characteristic for which cell?

- 1 mastcell
- 2 plasmacell
- 3 kupfer
- 4 Adipocyte



	Histology		
30	structure of framework of spleen and lymphnodes are composed of which fibers?		
	1	reticular	
	2	elastic	
	3	elaunin	
	4	collagen bundles	

	Histology		
31	which one is dense regular connective tissue?		
	1	Dermis of skin	
	2	Tendon ✓	
	3	Adipose tissue	
	4	Areolar tissue	

	Histology		
32	heat production is the most important function in which tissue?		
	1	tendon	
	2	dermis	
	3	brown adipose tissue ✓	
	4	areolar tissue	

	Histology		
33	Red blood cell is because it has In the wall.		
	1	fragile-actin	
	2	flexible-fibrinogen ✓	
	3	flexible-spectrin	
	4	fragile-spectrin	

	Histology		
34	which cell has fewest number in the normal blood smear?		
	1	Neutrophil	
	2	Basophil ✓	
	3	monocyte	
	4	Eosinophil	

	Histology		
35	in which cell we have major basic protein?		
	1	lymphocyte	
	2	monocyte ✓	
	3	macrophage	
	4	Eosinophil	

	Histology		
36	following diapedesis only type of leukocytes can return from the tissues back to the blood. Which one is this?		
	1	Lymphocyte ✓	
	2	monocyte	
	3	basophil	
	4	neutrophil	

Embryology: Dr. Amidi

	Embryology	Question time (seconds): →	
37	Which of the following develops from the multinucleated syncytium, the syncytiotrophoblast and serves as the beginnings of gas exchange between mother and embryo?		
	1	Bilaminar disk	
	2	Cytotrophoblast	
	3	Amniotic cavity	
	4	Lacunae	

	Embryology	Question time (seconds): →	
38	The amniotic cavity forms during the process of implantation of the blastocyst. The amniotic cavity forms within which of the following structures?		
	1	Epiblast ✓	
	2	Cytotrophoblast	
	3	Syncytiotrophoblast	
	4	Maternal endometrium	

	Embryology	Cell histology
39 C	In a growing embryo, the vertebrae encase notochord and neural tube in a solid bone like structure. Vertebrae are formed from which of the following embryological structures?	
	1	Dermatomes
	2	Myotomes
	3	Sclerotomes
	4	Primaxial domain

	Embryology	Question time (seconds): →
40 C	What structure has to degenerate so that implantation can occur?	
	1	Syncytiotrophoblast
	2	Functional layer of the endometrium
	3	Cytotrophoblast
	4	Zona Pellucida

	Embryology	Question time (seconds): →
41 C	Chemotherapy treatment often targets mitotically active cells. Although this treatment is beneficial for cancer, it can negatively effect implantation and embryonic growth. Which of the following layers would be most affected by a slowed cell division rate?	
	1	Cytotrophoblast
	2	Epiblast
	3	Amnioblast
	4	Syncytiotrophoblast

	Embryology	Question time (seconds): →
42 C	Extraembryonic somatic mesoderm is in close association with:	
	1	syncytiotrophoblast
	2	cytotrophoblast & amnion
	3	yolk sac
	4	syncytio- & cytotrophoblast

	Embryology	Question time (seconds): →
43	The definitive yolk sac of the embryo appears by what day?	
	1 13th	
	2 26th	
	3 72nd	
	4 94th	

	Embryology	Question time (seconds): →
44	Which of the following types of germ cells does not undergo cell division?	
	1 spermatogonia	
	2 primary oocytes	
	3 spermatids	
	4 secondary oocytes	

	Embryology	Question time (seconds): →
45	Spermatogonia, derived from primordial germ cells, divide by mitosis during:	
	1 postnatal periods	
	2 prenatal periods	
	3 the reproductive period	
	4 puberty	

	Embryology	Question time (seconds): →
46	4-year-old healthy female presents to your office to establish primary care. What phase of meiosis should her oocytes be in?	
	1 Meiosis I - prophase	
	2 Meiosis II - prophase	
	3 Meiosis I - metaphase	
	4 Meiosis II - metaphase	

	Embryology	Question time (seconds): →
47	Capacitation of spermatozoa is an activation process that involves changes in the surface coat and plasma membrane over the _____.	
	1 tail	
	2 body	
	3 neck	
	4 acrosome	

	Embryology	Question time (seconds): →
48 ✓	The early stages of cleavage are characterized by:	
	1 formation of a hollow ball of cells	
	2 formation of the zona pellucida	
	3 increase in the size of the cells in the zygote	
	4 increase in the number of cells in the zygote	

	Embryology	Question time (seconds): →
49	The LH surge:	
	1 results in ovulation	
	2 triggers resumption of meiosis within the oocyte	
	3 is caused by a positive feedback effect of estrogen	
	4 all of the above	

	Embryology	Question time (seconds): →
50	The primitive streak first appears at the beginning of the ____ week.	
	1 first	
	2 second	
	3 third	
	4 fourth	

	Embryology	Question time (seconds): →
51	Which of the following structures is believed to be a primary organizer or inducer during organogenesis?	
	1 somites	
	2 notochord	
	3 metanephric blastema	
	4 lens placode	

	Embryology	Question time (seconds): →
52	Cells from the primitive streak DO NOT become:	
	1 endoderm	
	2 intermediate mesoderm	
	3 paraxial mesoderm	
	4 amnioblasts	

	Embryology	Question time (seconds):	
	→		
53	Lanugo covers the body, some head hairs show, and fetal movements are felt by the mother when the fetus is about how many months old?		
	1	5	
	2	6	
	3	7	
	4	8	

	Embryology	Question time (seconds):	
	→		
54	During development, the notochordal process:		
	1	arises from involuting endodermal cells	
	2	extends from the prochordal plate to the primitive node	
	3	is involved in the induction of the primitive gut	
	4	becomes the appendicular skeleton	

	Embryology	Question time (seconds):	
	→		
55	Chorionic villi are designated as secondary chorionic villi when they:		
	1	contact the decidua basalis	
	2	are covered by syncytiotrophoblast	
	3	develop a mesenchymal core	
	4	give rise to branch villi	

	Embryology	Question time (seconds):	
	→		
56	The most important region of the decidua for the nourishment of the conceptus is the decidua _____.		
	1	parietalis	
	2	frondosum	
	3	capsularis	
	4	basalis	

	Embryology	Question time (seconds): →
57	Erythroblastosis fetalis occurs due to what?	
	1 AB incompatibility	
	2 Rh+ baby born to Rh- mother	
	3 In utero Rubella infection	
	4 In utero HIV infection	

	Embryology	Question time (seconds): →
58	What abnormality causes oligohydramnios?	
	1 esophageal atresia	
	2 anular pancreas	
	3 microcephaly	
	4 renal agenesis	

	Embryology	Question time (seconds): →
59	Somites differentiate to form all of the following, except	
	1 Notochord	
	2 Sclerotome	
	3 Myotome	
	4 Dermatome	

	Embryology	Question time (seconds): →
60	Pregnancy is maintained till term by	
	1 Follicle stimulating hormone	
	2 Luteinizing hormone	
	3 Progesterone	
	4 Prostaglandins	
	Embryology	Question time (seconds): →
61	In which stage of spermatogenesis, the chromosome number is halved?	
	1 Spermatogonium to primary spermatocyte	
	2 Primary spermatocyte to secondary spermatocyte	
	3 Secondary spermatocyte to spermatid	
	4 Spermatid to sperm	

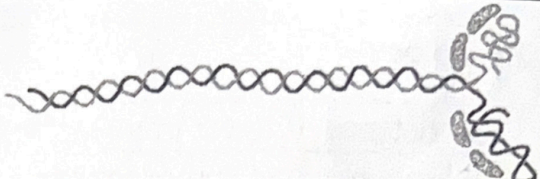
	Embryology	Question time (seconds): →	
62	Process/processes which sperms must undergo to be able to fertilize an oocyte are		
	1 Capacitation		
	2 Acrosome reaction		
	3 Both capacitation and acrosome reaction		
	4 Zonal reaction		

	Embryology	Question time (seconds): →	
63	In early pregnancy (i.e., upto 2 weeks), the placental barrier consists of all the structures, except		
	1 Cytotrophoblast		
	2 Syncytiotrophoblast		
	3 Extraembryonic mesoderm		
	4 Endothelium of fetal blood vessels		

Physiology: Dr. Ranjbaran

	Physiology	Question time (seconds): →	
64	In a skeletal muscle, is/are considered to be a functional unit.		
	1 myofibril		
	2 actin and myosin ✓		
	3 sarcomere		
	4 muscle fiber ✓		

	Physiology	Question time (seconds): →	
65	This molecule in one end is attached to the Z disks and in the other end is attached to the myosin filaments.		
	1 Actin		
	2 Titin ✓		
	3 Nebulin		
	4 Distrophin		

	Physiology	Question time (seconds): →	
66 C	What does this figure show? 		
	1 A myosin filament		
	2 A myosin molecule		
	3 F-actin strands		
	4 An actin filament		

	Physiology	Question time (seconds): →	
67 C	Which factor is considered to activate actin filaments in a skeletal muscle?		
	1 Myosin filaments		
	2 Ca ions		
	3 Tropomyosin-troponin complex		
	4 Mg ions		

	Physiology	Question time (seconds): →	
68	In which step of a contraction cycle, ADP and P are released from the myosin filaments?		
	1 Between steps 1 and 2		
	2 Between steps 2 and 3		
	3 Between steps 3 and 4		
	4 Between steps 4 and 1		

	Physiology	Question time (seconds): →	
69	In which step of a contraction cycle, A-M.ATP is seen?		
	1 Step 1		
	2 Step 2		
	3 Step 3		
	4 Step 4		

	Physiology	Question time (seconds): →	
70	The existence of fewer muscle fibers in a motor unit means		
	1 rapid reaction and exact control ✓		
	2 slow reaction and not require fine control		
	3 rapid reaction and not require fine control		
	4 slow reaction and exact control		

	Physiology	Question time (seconds): →	
71	In neuro-muscular junction, the membrane invagination of a muscle fiber is called		
	1 Synaptic gutter		
	2 Sub-neural cleft		
	3 Active zone		
	4 Dense bar		

	Physiology	Question time (seconds): →	
72	Comparing skeletal muscle with large nerves, which factor is considered to be the same in both?		
	1 Duration of the action potential		
	2 Resting Membrane Potential		
	3 Membrane permeability to Na ions		
	4 Velocity of Conduction		

	Physiology	Question time (seconds): →	
73	Which of the following mediate(s) excitation-contraction coupling in a skeletal muscle?		
	1 Terminal cisternae		
	2 Longitudinal tubules		
	3 Dihydropyridine- ryanodine receptors		
	4 Troponin complex		

	Physiology	Question time (seconds): →	
74	The main mechanism by which acetyl choline molecules are removed from synaptic cleft is		
	1 Diffusion to the out of the synaptic space		
	2 Reabsorbing to the nerve ending		
	3 Entering the blood stream		
	4 Destroying by the enzyme acetylcholinesterase ✓		

	Physiology	Question time (seconds): →	
75	In neuro-muscular junction, calcium entry into the nerve ending following an action potential leads to		
	1 binding to troponin C and making contraction		
	2 phosphorylation of synapsin and transmitter release		
	3 opening of voltage gated na channels ✓		
	4 hyperpolarization		

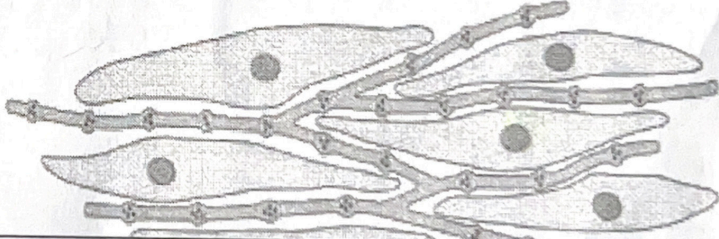
	Physiology	Question time (seconds): →	
76	Which of the following refers to the properties of electrical synapses?		
	1 No cytoplasmic continuity		
	2 Existence of synaptic delay		
	3 Unidirectional transmission		
	4 Synchronous signaling ✓		

	Physiology	Question time (seconds): →	
77	Which of the following refers to the difference between skeletal and smooth muscles?		
	1 Existence of actin and myosin filaments		
	2 Ca ions required for contraction		
	3 ATP hydrolysis for contraction		
	4 Duration of contraction ✓		

COUNTINUE TO THE NEXT PAGE....

	Physiology	Question time (seconds): →	
78	Which of the following best describes the cause of slow cycling of the myosin cross-bridges in the smooth muscle fibers?		
	1 Tropomyosin coverage on actin filaments		
	2 Second messenger activation makes it long.		
	3 Far less ATPase activity of myosin cross bridges		
	4 Slow activity of troponin C		

	Physiology	Question time (seconds): →	
79	Dephosphorylation of myosin light chains must occur for		
	1 smooth muscle relaxation		
	2 skeletal muscle relaxation		
	3 smooth muscle contraction		
	4 skeletal muscle contraction		

	Physiology	Question time (seconds): →	
80	What does this figure show?		
			
	1 Neuromuscular junctions in unitary smooth muscle		
	2 Neuromuscular junctions in multi-unit smooth muscle		
	3 Neuromuscular junctions in skeletal muscle		
	4 Release of transmitters no matter how to innervate		